

Week 2 Project Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Course Name: ECE501 Digital Image Processing (DIP)

Group Number: Group 15

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Week Number: Week 2

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Objectives of the Week

- Finalise the dataset for the project and acquire it locally in a reproducible folder layout.
- Understand the structure of the AITEX Fabric Image Database — fabric types, defect types, image dimensions, and the format of the supplied ground-truth masks.
- Implement a working pre-processing stage (grayscale handling, patch extraction, noise filtering, contrast and illumination normalisation) as the front end of the pipeline.
- Fix the patch-based processing strategy that the texture descriptors in the coming weeks will operate on.
- Keep the workload realistic given that the Mid-Semester Examination Period begins on 19 September 2026.

Work Completed This Week

A. Dataset selection and exploration

- Finalised the **AITEX Fabric Image Database (AFID)** as the project dataset, in line with the project definition issued for this problem statement. The earlier Week 1 shortlist (AITEX/TILDA-style options) has therefore been closed out.
- Downloaded the database and organised it under `data/raw/`, with defect-free images, defective images and the corresponding binary ground-truth masks kept in separate sub-directories.
- Confirmed the composition of the database: 245 grayscale images of size 4096×256 pixels spanning 7 fabric structures, of which 140 are defect-free (20 per fabric type) and 105 contain defects drawn from 12 defect classes commonly seen in the textile industry (broken end, broken pick, broken yarn, weft curling, fuzzy ball, cut selvage, crease, warp ball, knot, contamination, nep and weft crack).
- Verified that every defective image has a paired pixel-level mask, and wrote a small script to catalogue the image count per fabric type and per defect type into a CSV summary.
- Noted the unusual 16:1 aspect ratio of the images and decided to process them as **256×256 patches** extracted along the strip, which matches the region-of-interest size used in the database documentation and gives a natural unit for localisation.

B. Pre-processing implementation

- Wrote a pre-processing module in Python using OpenCV and NumPy, exposed as a single callable stage so that later descriptor experiments can switch individual steps on and off.
- Implemented: grayscale loading and dtype normalisation; sliding-window patch extraction with a configurable stride; resizing utilities; median and Gaussian filtering for sensor noise; and contrast normalisation using both global histogram equalisation and CLAHE.

- Implemented mask handling to match: masks are binarised and tiled with the same window and stride as the images, so that each image patch carries its own ground-truth patch.
- Ran the stage visually on a handful of defective samples from different fabric types and compared raw, filtered and contrast-normalised versions side by side to check that visible defects survive the normalisation.

C. Scope note

The Mid-Semester Examination Period (19–27 September 2026) falls immediately after this reporting week, so effort was deliberately concentrated on dataset acquisition and the pre-processing front end rather than on descriptor implementation. Texture descriptor work has been carried forward to the next reporting period.

Literature Review

Topic/Paper	Key Takeaway	Relevance to the Project
AFID: A Public Fabric Image Database for Defect Detection ¹	Describes the construction of the AITEX database, the imaging setup used in the factory environment, the 12 defect categories, and the rationale for a common public benchmark in textile inspection.	The primary reference for our dataset. Fixes the image geometry, the defect vocabulary and the mask format our pipeline must consume, and gives the baseline against which our results will be positioned.
Contrast Limited Adaptive Histogram Equalization (CLAHE) ²	Equalises contrast over local tiles with a clip limit that bounds noise amplification, rather than applying one global transformation to the whole image.	Chosen for the pre-processing stage because fabric strips show uneven illumination along their length; the clip limit is what keeps low-contrast defects from being flattened into the background.
Multiresolution Gray-Scale and Rotation Invariant Texture Classification with Local Binary Patterns ³	Generalises LBP to multiple radii and neighbourhood sizes, and introduces the uniform-pattern and rotation-invariant variants that compress the histogram while retaining discriminative power.	Directly informs the 'adaptive multi-scale' requirement of the project: it gives us the radius/neighbour pairs to sweep when we compare fixed against adaptive descriptor parameters.
A Threshold Selection Method from Gray-Level Histograms ⁴	Selects a threshold by maximising between-class variance in the intensity histogram, with no parameter tuning required.	Candidate for converting the texture-deviation map into a binary defect mask in the detection stage, and a sensible unsupervised baseline before any adaptive or local thresholding is introduced.

Challenges Faced

- The Mid-Semester Examination Period restricted the hands-on time available, so descriptor implementation had to be deferred by one week.
- The 16:1 aspect ratio of the images forced an early decision on patch size and stride. A large window gives a more stable texture estimate but coarser localisation, while a small window localises better but makes the texture statistics noisy — we have fixed 256×256 for now and will treat the window size as a parameter to sweep.

- Class imbalance is significant: 105 defective images against 140 defect-free ones, and those 105 are spread across 12 defect types, so some defect categories will have very few samples. This will make the defect-type-wise performance analysis required by the project statement statistically thin for the rarer classes.
- Contrast normalisation is a double-edged step — aggressive equalisation improves visibility of some defects while washing out subtle low-contrast ones such as neps and slight creases. Choosing the CLAHE clip limit is therefore not a cosmetic decision.
- Still open: whether the 'normal texture' reference should be modelled separately per fabric type or once globally. Per-type modelling is likely more accurate but reduces the amount of reference data available for each model.

Plan for Next Week

- Implement GLCM feature extraction (contrast, correlation, energy, homogeneity) over the patch grid and inspect how the statistics separate defective from defect-free patches.
- Implement LBP histograms over the same patch grid, starting with the uniform rotation-invariant variant at one or two radii.
- Build a reference model of normal fabric texture from the defect-free images and produce the first texture-deviation maps for a small test subset.
- Set up the evaluation harness — pixel-level and image-level IoU, precision, recall and F1-score against the supplied ground-truth masks — so that every later experiment reports against a fixed protocol.
- Align the GitHub repository with the required naming convention and the prescribed directories (Mid_Sem_Report, End_Sem_Report, Results, Codes), and commit the pre-processing code alongside this report.

Conclusion

Week 2 moved the project from a defined scope to a concrete data foundation. The dataset is now fixed and understood, its structure has been catalogued, and a working pre-processing stage produces normalised 256×256 patches with matched ground-truth masks — which is exactly the input the texture descriptors need. Progress was intentionally kept to this front-end work because the Mid-Semester Examination Period begins on 19 September; descriptor implementation and the first deviation maps are the priority for the next reporting period.

Citations

- 1 J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A Public Fabric Database for Defect Detection Methods and Results," *AUTEX Research Journal*, vol. 19, no. 4, pp. 363–374, 2019.
- 2 K. Zuiderveld, "Contrast Limited Adaptive Histogram Equalization," in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA, USA: Academic Press, 1994, pp. 474–485.
- 3 T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution Gray-Scale and Rotation Invariant Texture Classification with Local Binary Patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 7, pp. 971–987, 2002.
- 4 N. Otsu, "A Threshold Selection Method from Gray-Level Histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.